

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

NAME: A SHIVA SHANKAR REDDY

REG NO:192425174

COURSE OUTCOME COVERED

CO5: CO5 – AT3 Mock Interview

EVAULATION:

Claude

New

Projects

Artifacts

Code

Customize

Chats and tasks

Following file instructions carefully

Q/I Knapsack algorithm for server

Secure password pattern validation

Hierarchical reinforcement learning

Hierarchical reinforcement learning

Q/I Knapsack optimization using dynamic programming

Document answers with formulas and LaTeX

Professional Q&A document format

Claude's functionality with and without code

File upload and document completion

Federated proximal for stock market

Claude's advantages over other AI models

Design

Shiva - Free

Following file instructions carefully

Question-wise Performance

Q#	Topic	Performance
1	Backtracking vs Brute Force	Correct, concise
2	State-Space Tree (N-Queens)	Correct
3	Pruning/Bounding Function	Correct
4	N-Queens Constraints	Correct
5	Subset Sum	Correct
6	Hamiltonian Cycle & Complexity Class	Correct
7	NP vs NP-Complete & Reduction	Correct, good depth
8	Graph Coloring & Application	Correct, good example
9	Exact vs Approximation	Correct
10	Scenario: Greedy Coloring for Large Graphs	Correct, well-reasoned

Technical Strengths

- Solid grasp of backtracking, pruning, and state-space tree concepts.
- Correctly distinguished NP/NP-Complete, and the role of polynomial-time reduction — a commonly confused topic.
- Applied concepts to real-world scenarios (exam scheduling, greedy coloring for large graphs) accurately.
- Consistent, correct answers across all 10 questions with no major technical errors.

Areas Requiring Improvement

- Answers were often correct but quite brief — expanding with a small example (e.g., walking through N-Queens for a 4x4 board, or a small Subset Sum instance) would demonstrate deeper mastery.
- Could elaborate more on why certain complexities arise (e.g., time complexity of backtracking algorithms in Big-O terms) rather than just describing behavior.
- More elaboration on trade-offs (e.g., specific approximation ratios, or naming known heuristics like Welsh-Powell for graph coloring) would strengthen analytical depth.

+ Write a message...

Claude is AI and can make mistakes. Please double-check responses.

Sonnet 5 - Low

33°C Mostly cloudy

Search

10:38 31-08-2026

Claude

New

Projects

Artifacts

Code

Customize

Chats and tasks

Following file instructions carefully

Q/I Knapsack algorithm for server

Secure password pattern validation

Hierarchical reinforcement learning

Hierarchical reinforcement learning

Q/I Knapsack optimization using dynamic programming

Document answers with formulas and LaTeX

Professional Q&A document format

Claude's functionality with and without code

File upload and document completion

Federated proximal for stock market

Claude's advantages over other AI models

Design

Shiva - Free

Following file instructions carefully

- Could elaborate more on why certain complexities arise (e.g., time complexity of backtracking algorithms in Big-O terms) rather than just describing behavior.
- More elaboration on trade-offs (e.g., specific approximation ratios, or naming known heuristics like Welsh-Powell for graph coloring) would strengthen analytical depth.

Rubric-Based Evaluation

Criterion	Level Achieved	Marks Obtained	Weightage
Technical Knowledge	Level 4	22/25	25
Problem Solving	Level 3	20/25	25
Analytical Thinking	Level 3	11/15	15
Communication	Level 3	11/15	15
Confidence	Level 4	9/10	10
Response to Questions	Level 4	9/10	10
Total	—	82/100	100

Overall Performance

Strong performance overall. Shiva demonstrated a clear and accurate conceptual understanding across backtracking, complexity theory (P/NP/NP-Complete), and greedy/approximation strategies. Answers were consistently correct, though brief — moving from "correct definitions" to "correct definitions + example + complexity justification" would push this into the Level 4 range across all criteria.

Suggestions for Improvement

- Practice explaining answers with a small worked example (e.g., trace through N-Queens or Subset Sum on paper) to show problem-solving depth, not just definitions.
- Get comfortable stating time complexities (e.g., $O(N!)$ for N-Queens backtracking) alongside conceptual answers.
- Learn names of a few standard heuristics/approximation algorithms (Welsh-Powell for coloring, greedy for Vertex Cover, etc.) to cite in scenario questions.
- Practice elaborating slightly longer answers (aim for the full 1-1.5 min per question) to fully showcase analytical thinking and communication.

+ Write a message...

Claude is AI and can make mistakes. Please double-check responses.

Sonnet 5 - Low

33°C Mostly cloudy

Search

10:37 31-08-2026